

Multiclass Classification

CIFAR10

airplane

automobile

bird

cat

deer

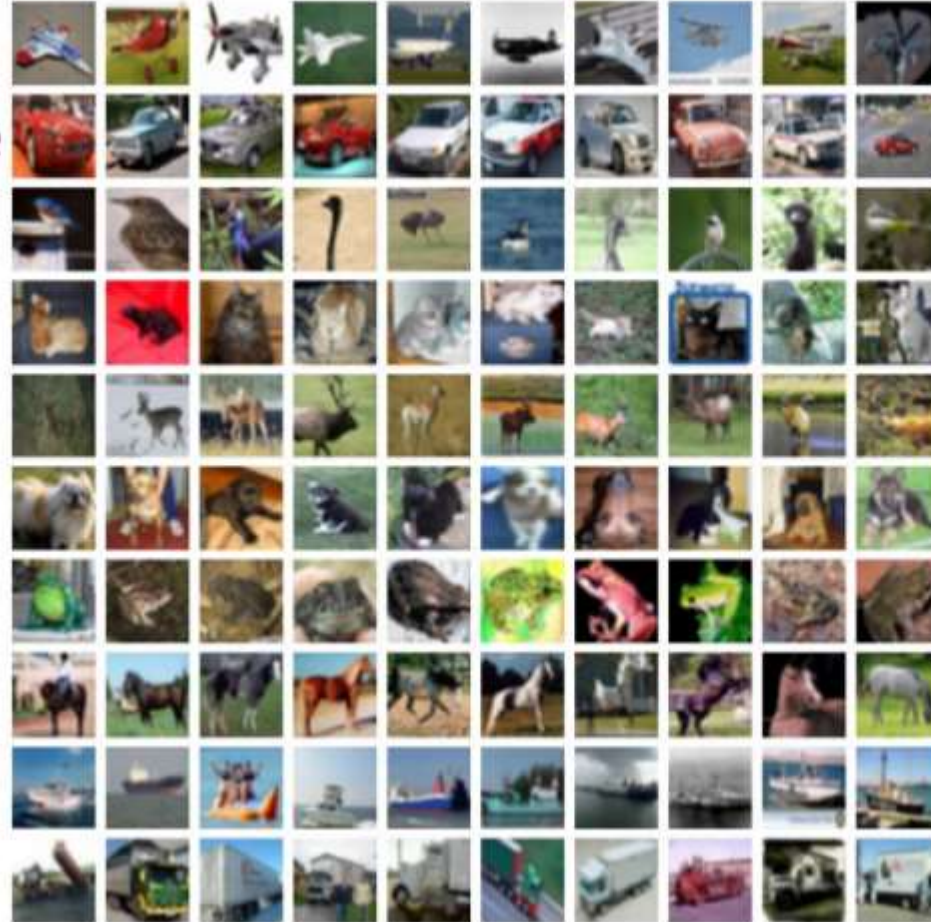
dog

frog

horse

ship

truck



50,000 training images
each image is **32x32x3**

10,000 test images.

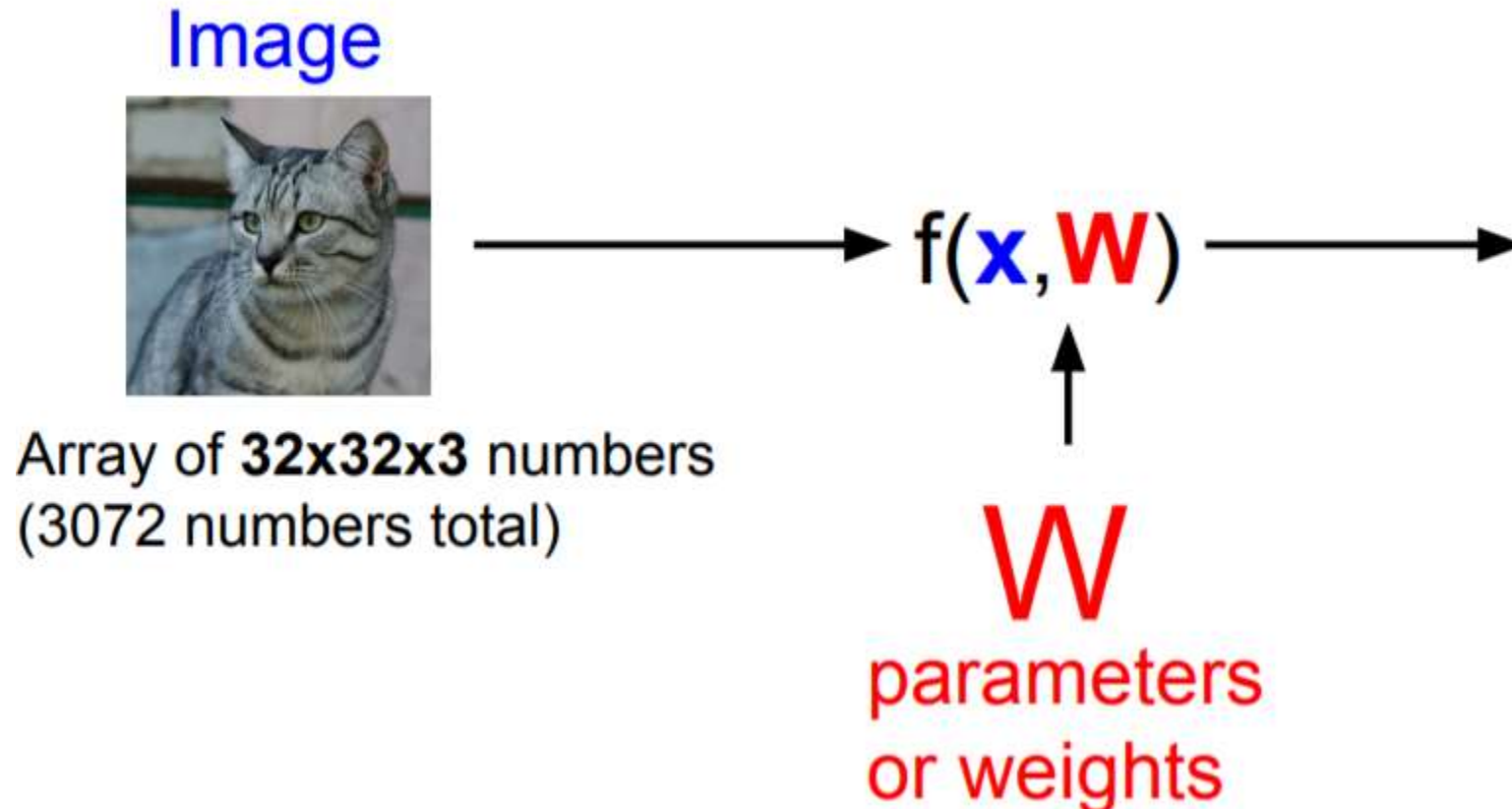
Parametric Approach

Image

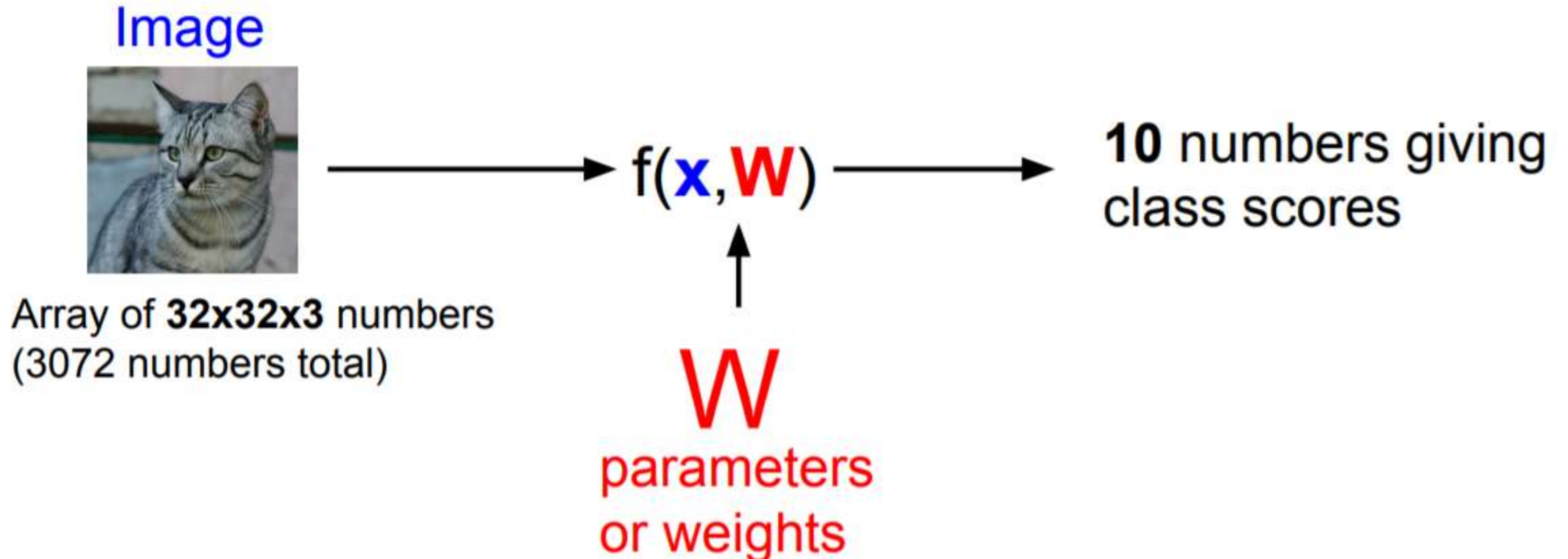


Array of **32x32x3** numbers
(3072 numbers total)

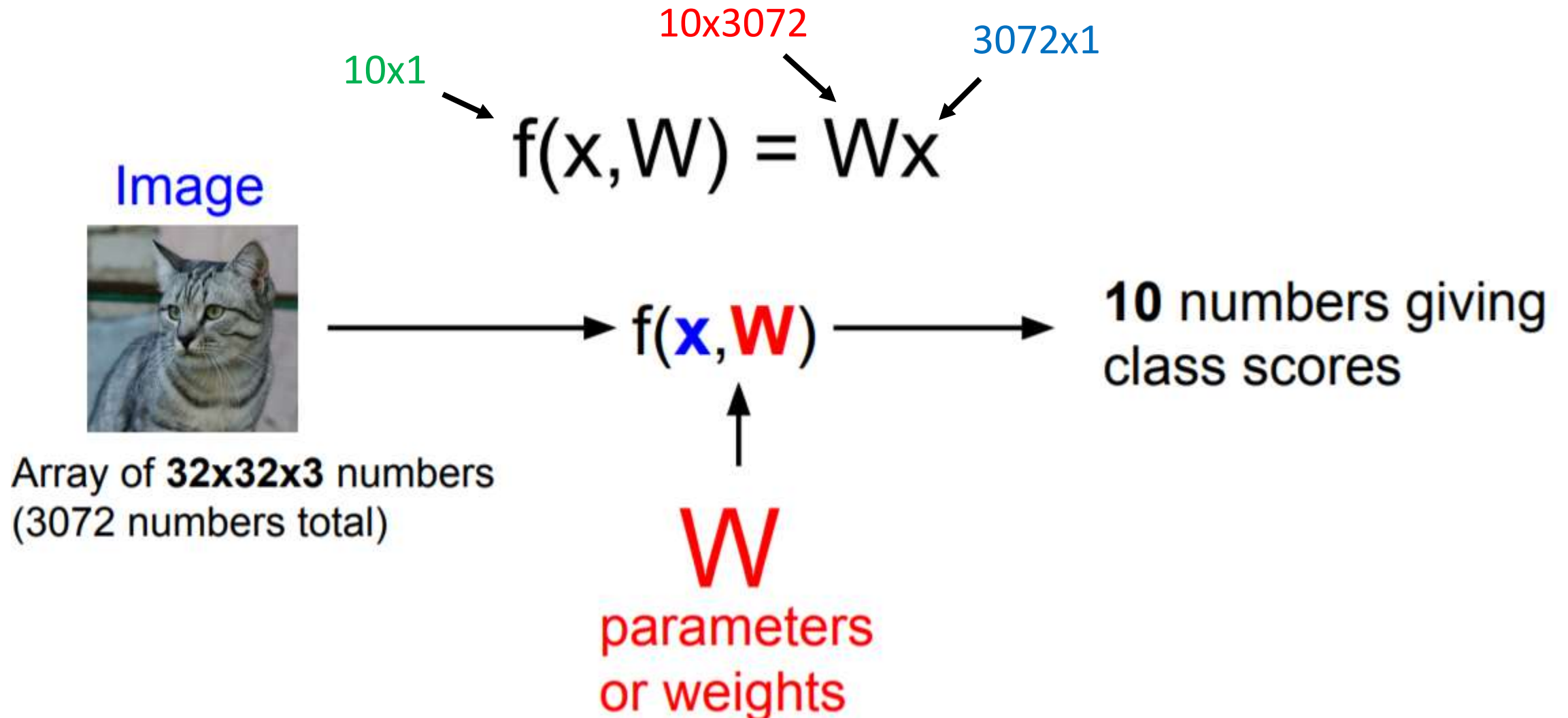
Parametric Approach



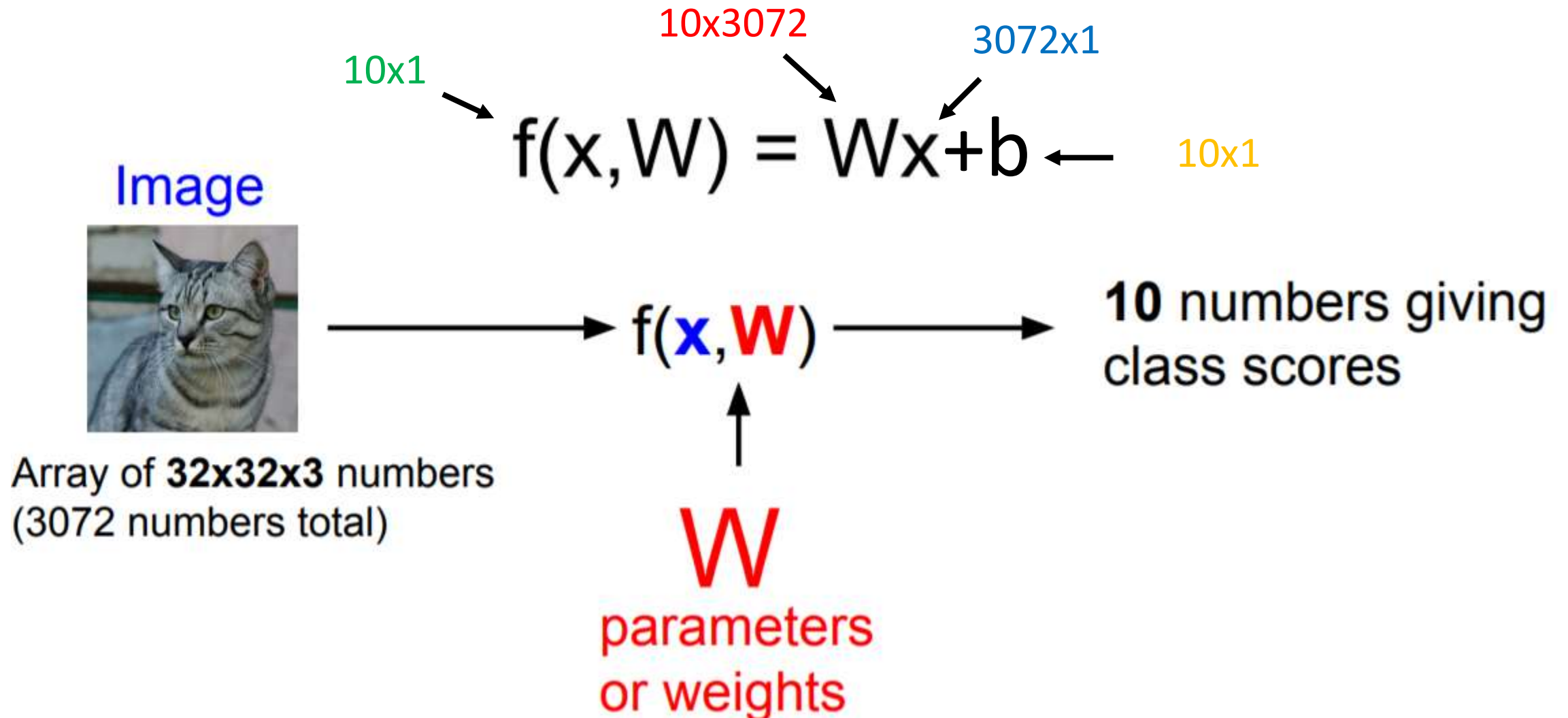
Parametric Approach



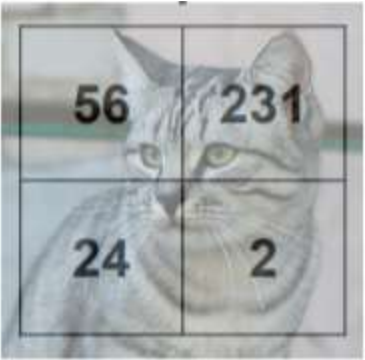
Parametric Approach: linear classifier



Parametric Approach: linear classifier

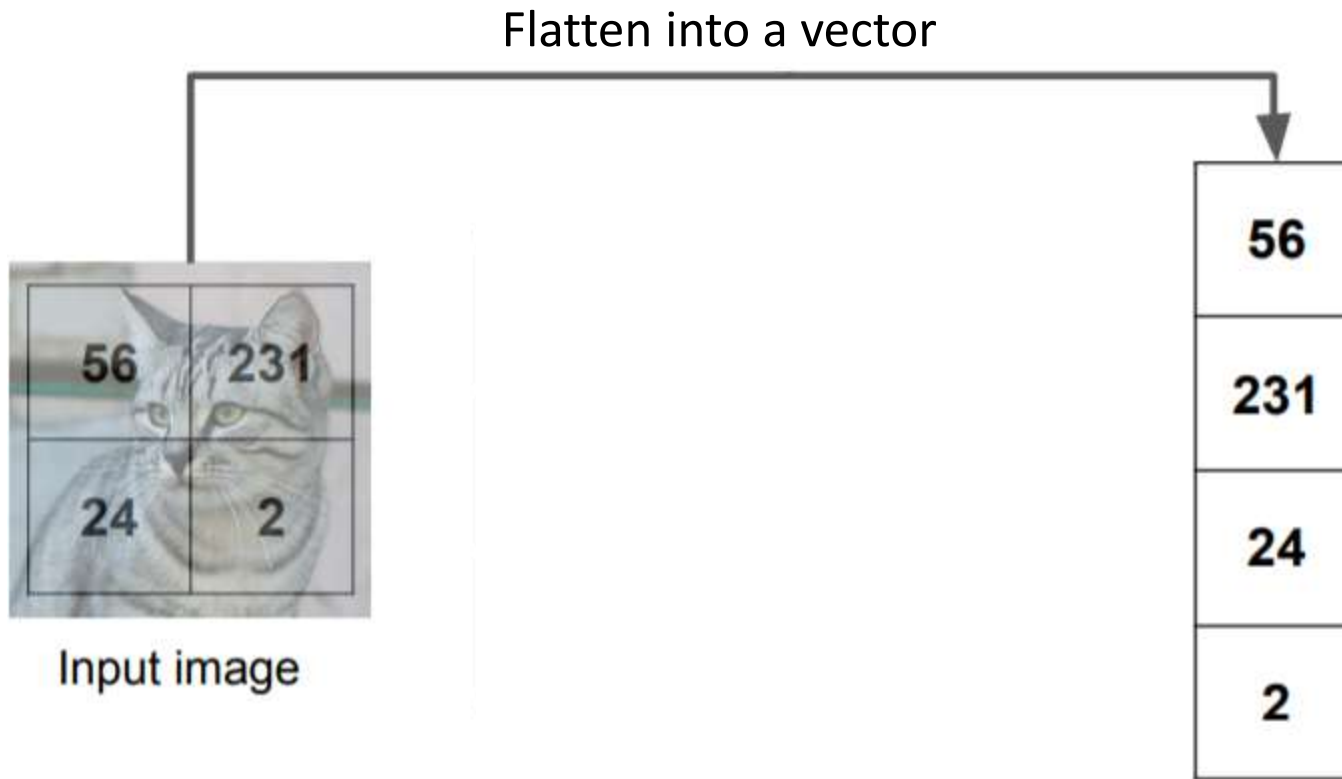


Example with an image with 4 pixels, and 3 classes (cat/dog/ship)

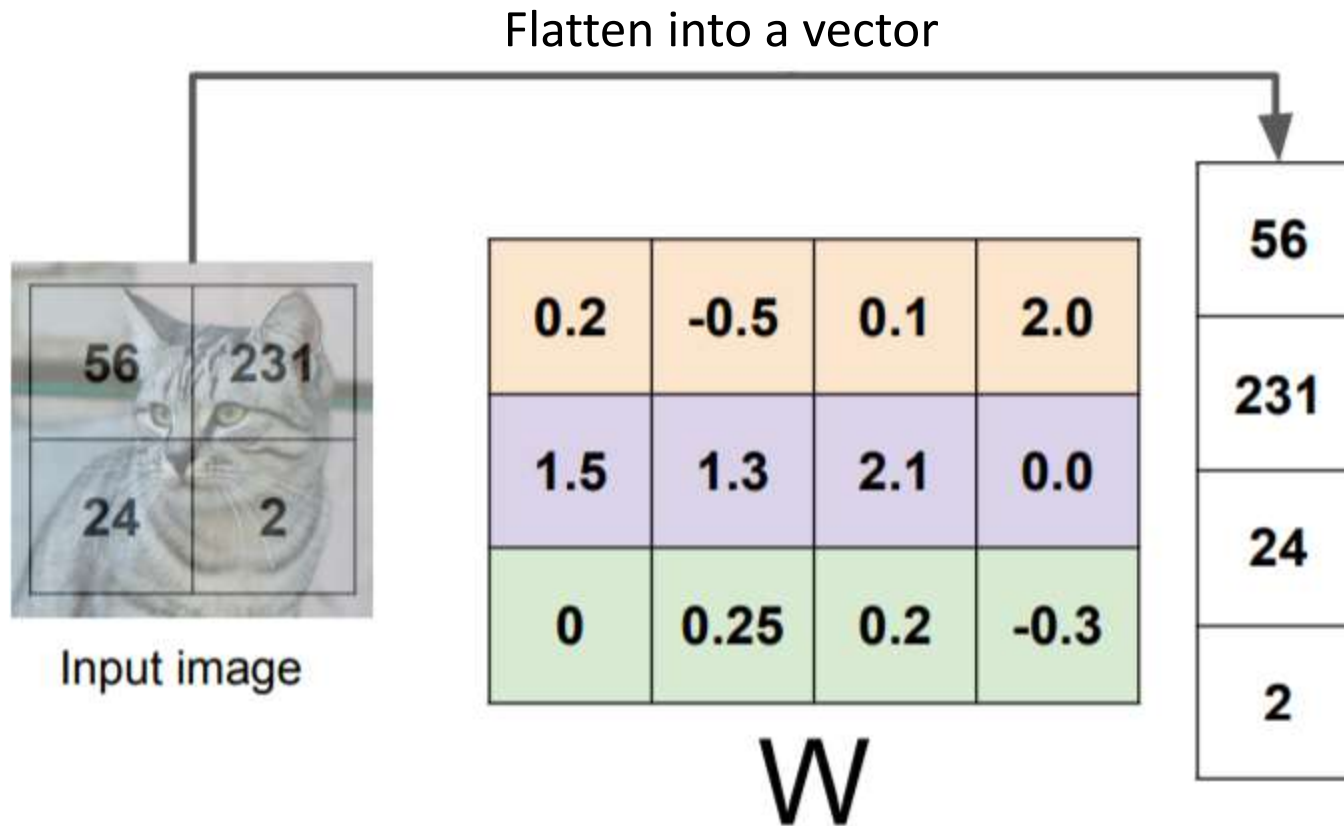


Input image

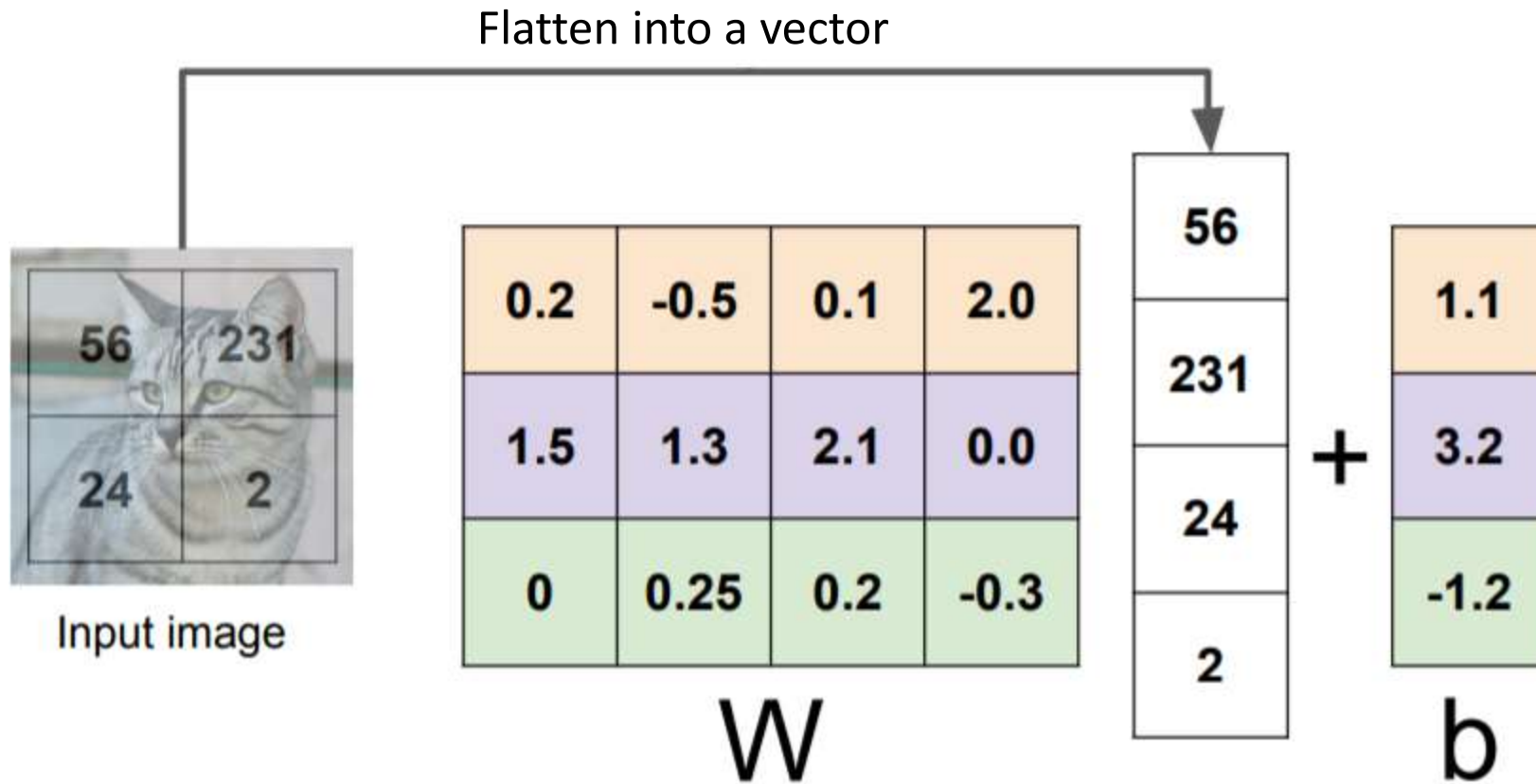
Example with an image with 4 pixels, and 3 classes (cat/dog/ship)



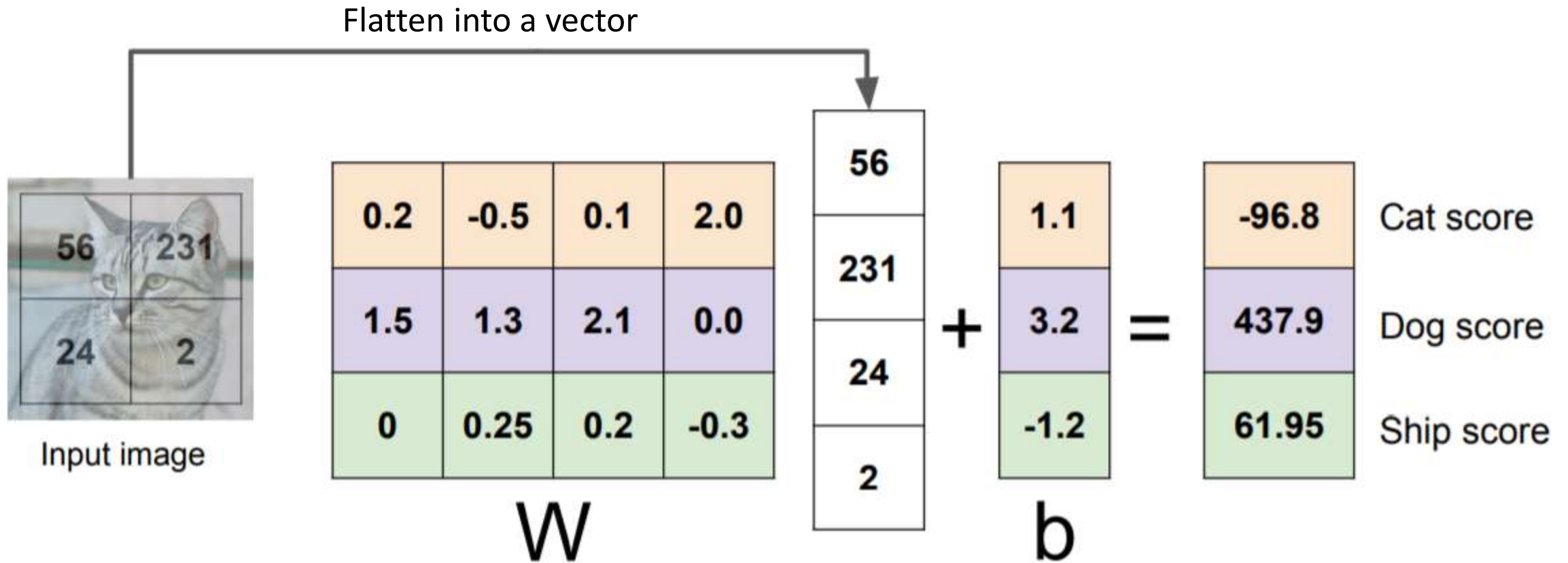
Example with an image with 4 pixels, and 3 classes (cat/dog/ship)



Example with an image with 4 pixels, and 3 classes (cat/dog/ship)



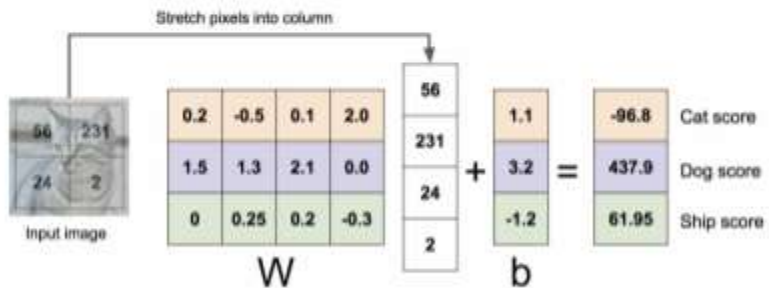
Example with an image with 4 pixels, and 3 classes (cat/dog/ship)



Linear Classifier: Three Viewpoints

Algebraic Viewpoint

$$f(x, W) = Wx$$



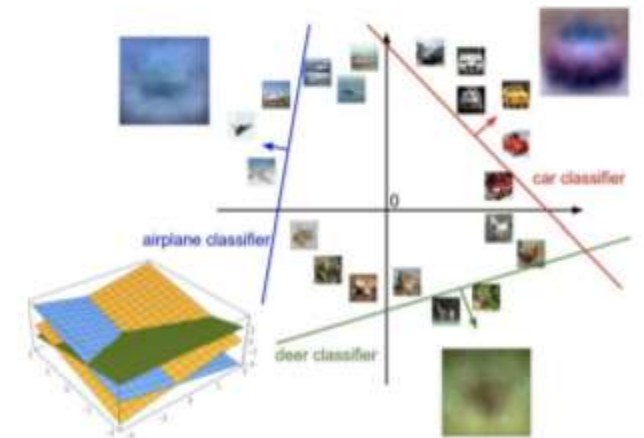
Visual Viewpoint

One template
per class



Geometric Viewpoint

Hyperplanes
cutting up space



Linear Classifier



airplane	-3.45	-0.51	3.42
automobile	-8.87	6.04	4.64
bird	0.09	5.31	2.65
cat	2.9	-4.22	5.1
deer	4.48	-4.19	2.64
dog	8.02	3.58	5.55
frog	3.78	4.49	-4.34
horse	1.06	-4.37	-1.5
ship	-0.36	-2.09	-4.79
truck	-0.72	-2.93	6.14

Cat image by Nikita is licensed under [CC-BY 2.0](#); Car image is [CC0 1.0](#) public domain; Frog image is in the public domain

Linear Classifier



airplane	-3.45	-0.51	3.42
automobile	-8.87	6.04	4.64
bird	0.09	5.31	2.65
cat	2.9	-4.22	5.1
deer	4.48	-4.19	2.64
dog	8.02	3.58	5.55
frog	3.78	4.49	-4.34
horse	1.06	-4.37	-1.5
ship	-0.36	-2.09	-4.79
truck	-0.72	-2.93	6.14

Cat image by Nikita is licensed under CC-BY 2.0; Car image is CC0 1.0 public domain; Frog image is in the public domain

TODO:

1. Define a **loss function** that quantifies our unhappiness with the scores across the training data.

Linear Classifier



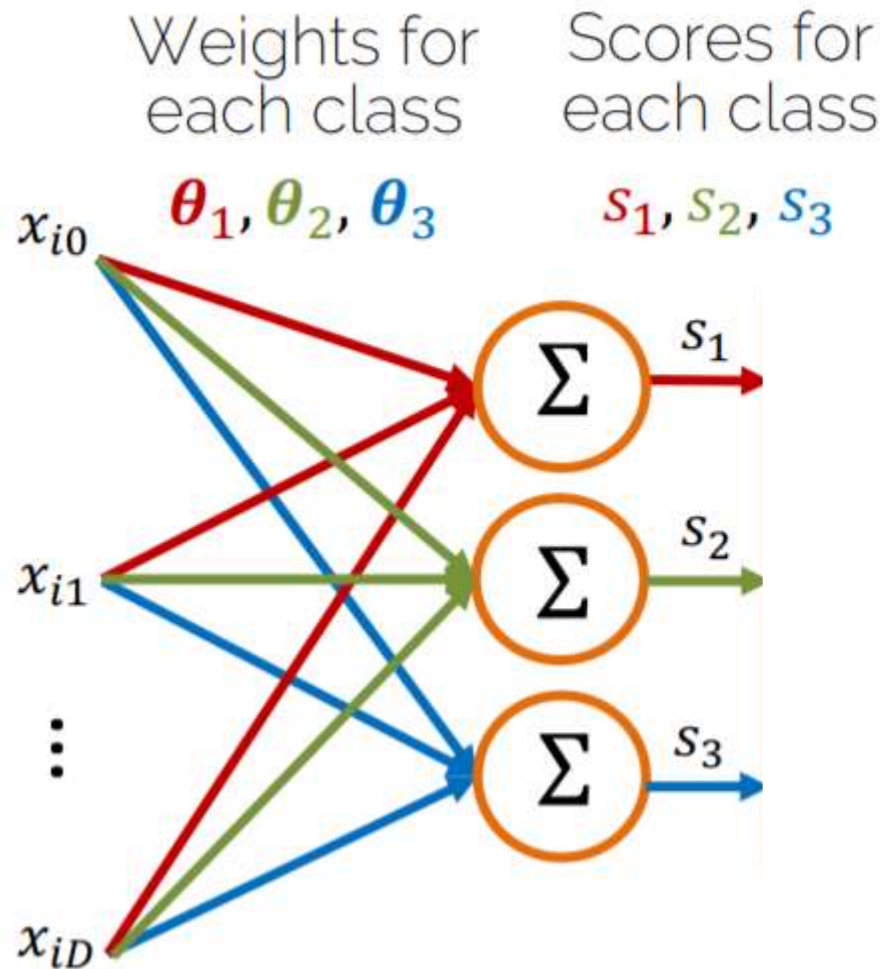
airplane	-3.45	-0.51	3.42
automobile	-8.87	6.04	4.64
bird	0.09	5.31	2.65
cat	2.9	-4.22	5.1
deer	4.48	-4.19	2.64
dog	8.02	3.58	5.55
frog	3.78	4.49	-4.34
horse	1.06	-4.37	-1.5
ship	-0.36	-2.09	-4.79
truck	-0.72	-2.93	6.14

Cat image by Nikita is licensed under CC-BY 2.0; Car image is CC0 1.0 public domain; Frog image is in the public domain

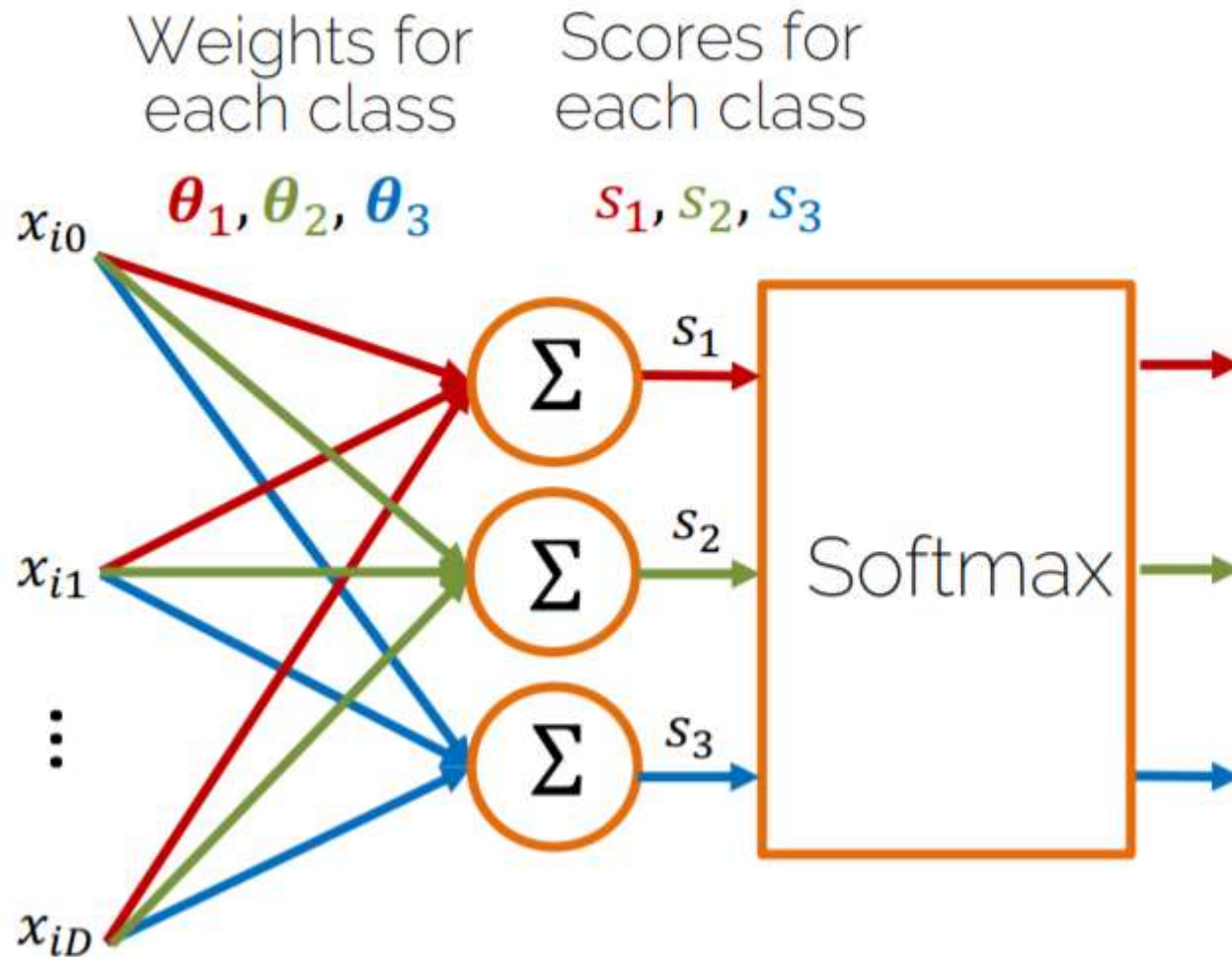
TODO:

1. Define a **loss function** that quantifies our unhappiness with the scores across the training data.
2. Come up with a way of efficiently finding the parameters that minimize the loss function.
(optimization)

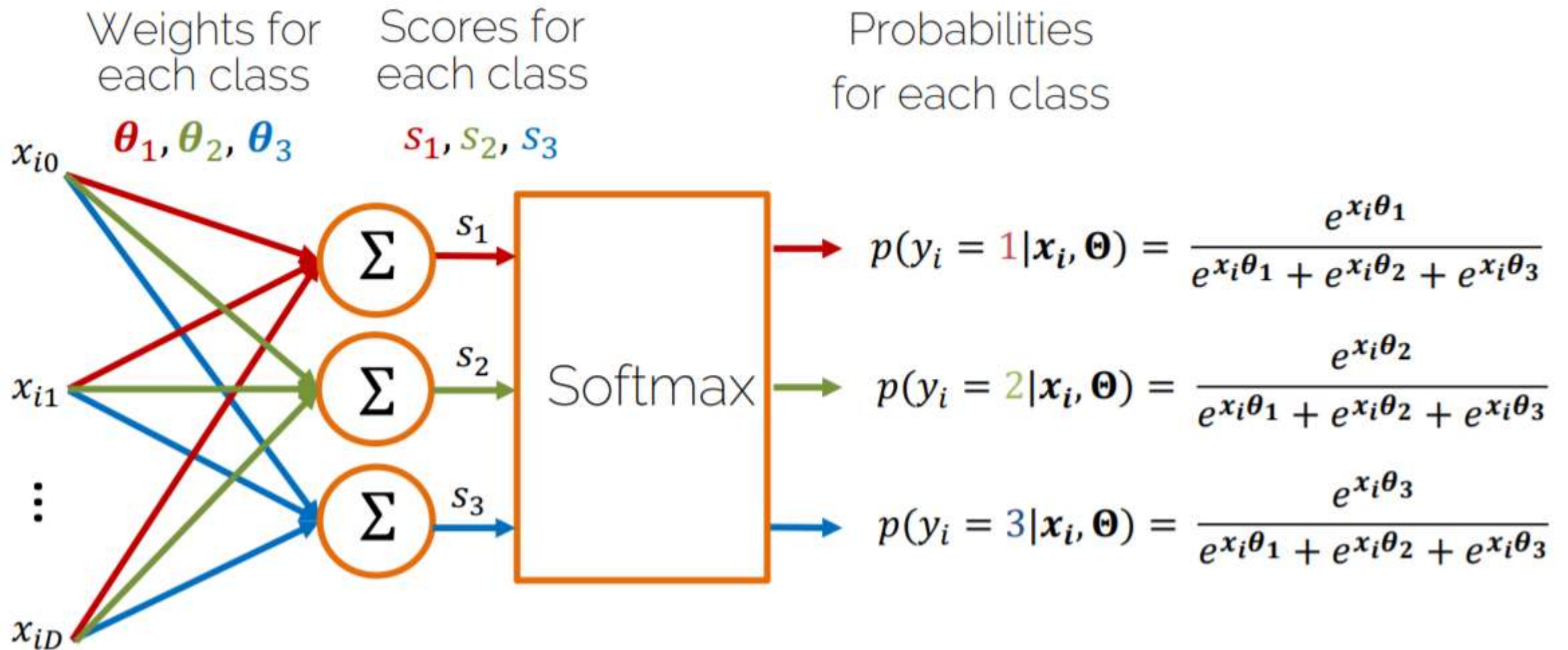
Multiclass Classification: Softmax



Multiclass Classification: Softmax



Multiclass Classification: Softmax



Multiclass Classification: Softmax

Probability of
the true class → $p(y_i | \mathbf{x}_i; \Theta)$

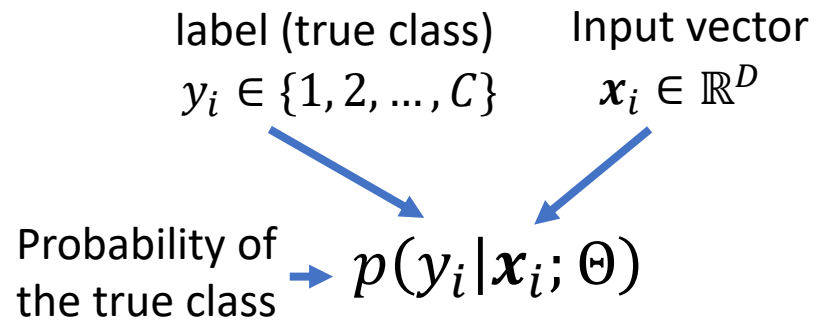
Multiclass Classification: Softmax

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

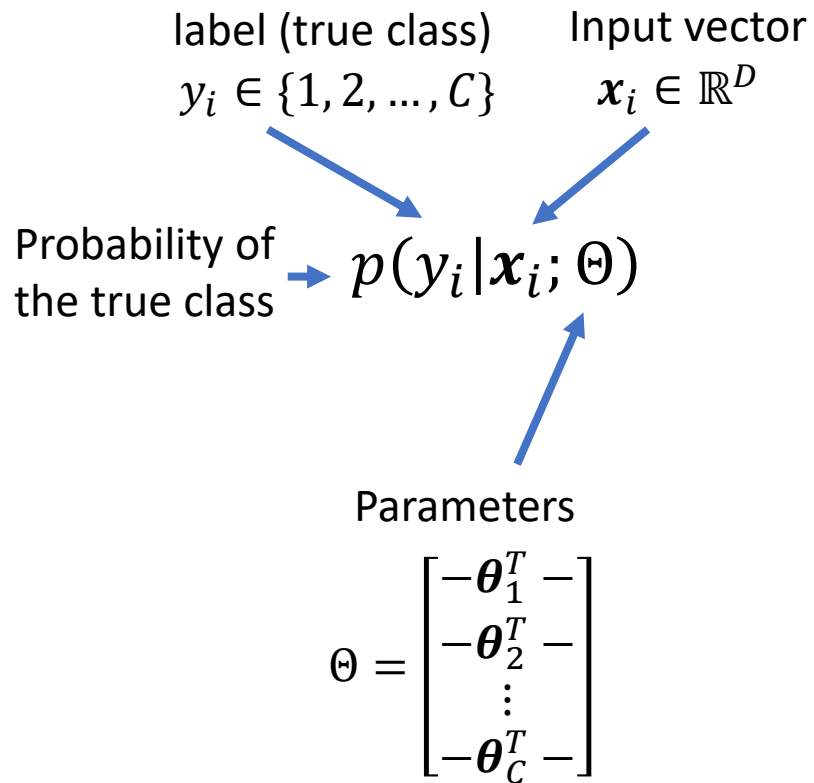


Probability of
the true class → $p(y_i | \mathbf{x}_i; \Theta)$

Multiclass Classification: Softmax



Multiclass Classification: Softmax



Multiclass Classification: Softmax

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Probability of
the true class $\rightarrow$

$$p(y_i | \mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Parameters

$$\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$$

The diagram illustrates the Softmax equation for multiclass classification. It shows three inputs contributing to the probability calculation: the label (true class) y_i , the input vector $\mathbf{x}_i$, and the parameters Θ . Arrows point from each of these inputs to the equation $p(y_i | \mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$. The label y_i is specified as belonging to the set $\{1, 2, \dots, C\}$. The input vector $\mathbf{x}_i$ is specified as belonging to $\mathbb{R}^D$. The parameters Θ are shown as a matrix with rows $[-\boldsymbol{\theta}_1^T, -]$, $[-\boldsymbol{\theta}_2^T, -]$, $[\vdots]$, and $[-\boldsymbol{\theta}_C^T, -]$. A blue arrow also points from the text 'Probability of the true class' to the left side of the equation.

Multiclass Classification: Softmax

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Probability of the true class $\rightarrow$

$p(y_i | \mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$

s : score of the class

Parameters

$$\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$$

The diagram illustrates the Softmax equation for multiclass classification. It features the equation $p(y_i | \mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$ in the center. Four blue arrows point to different parts of the equation: one from the label y_i to the subscript in the numerator, one from the input vector $\mathbf{x}_i$ to the argument of the probability function, one from the parameter matrix Θ to the argument of the probability function, and one from the text ' s : score of the class' to the s_{y_i} in the numerator. Above the equation, the label $y_i \in \{1, 2, \dots, C\}$ and the input vector $\mathbf{x}_i \in \mathbb{R}^D$ are defined. Below the equation, the parameter matrix Θ is shown as a matrix with rows $[-\boldsymbol{\theta}_1^T, -]$, $[-\boldsymbol{\theta}_2^T, -]$, $[\vdots]$, and $[-\boldsymbol{\theta}_C^T, -]$. To the left of the equation, the text 'Probability of the true class' is followed by an arrow pointing to the probability function $p(y_i | \mathbf{x}_i; \Theta)$.

Multiclass Classification: Softmax

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Probability of the true class $\rightarrow$

$p(y_i | \mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$

s : score of the class

C : number of classes

Parameters

$$\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$$

Multiclass Classification: Softmax

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Probability of the true class $\rightarrow p(y_i | \mathbf{x}_i; \Theta)$

Parameters
 $\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$

s : score of the class

C : number of classes

$$p(y_i | \mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}} = \frac{e^{\boldsymbol{\theta}_{y_i}^T \mathbf{x}_i}}{\sum_{k=1}^C e^{\boldsymbol{\theta}_k^T \mathbf{x}_i}}$$

Multiclass Classification: Softmax

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Probability of the true class $\rightarrow p(y_i | \mathbf{x}_i; \Theta)$

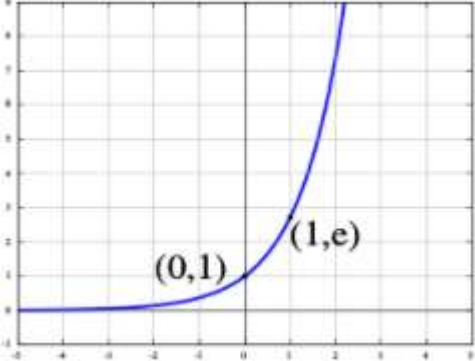
Parameters
 $\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$

s: score of the class

C: number of classes

$$p(y_i | \mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}} = \frac{e^{\boldsymbol{\theta}_{y_i}^T \mathbf{x}_i}}{\sum_{k=1}^C e^{\boldsymbol{\theta}_k^T \mathbf{x}_i}}$$

exponential function



Multiclass Classification: Softmax

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Probability of the true class $\rightarrow p(y_i | \mathbf{x}_i; \Theta)$

Parameters
 $\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$

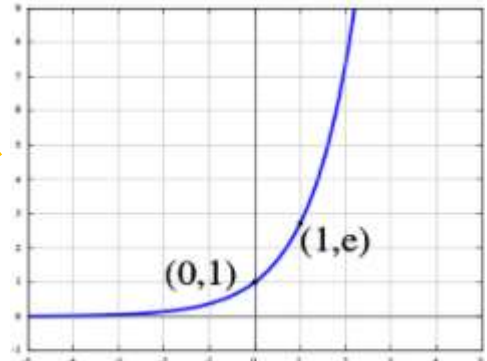
s : score of the class

C : number of classes

normalization

$$p(y_i | \mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}} = \frac{e^{\boldsymbol{\theta}_{y_i}^T \mathbf{x}_i}}{\sum_{k=1}^C e^{\boldsymbol{\theta}_k^T \mathbf{x}_i}}$$

exponential function



The exponential function makes sure probability > 0

Normalization makes sure probabilities sum up to 1.

Multiclass Classification: Softmax

Numerical stability

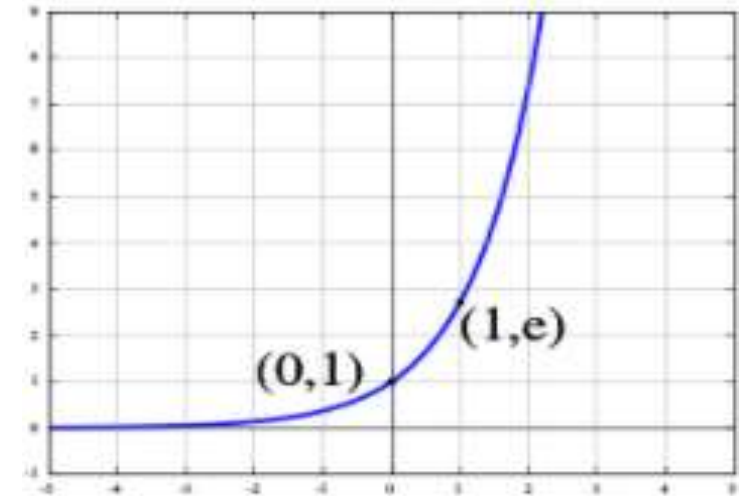
$$p(y_i|\mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Multiclass Classification: Softmax

Numerical stability

$$p(y_i|\mathbf{x}_i; \Theta) = \frac{e^{s_{yi}}}{\sum_{k=1}^C e^{s_k}}$$

exponential function

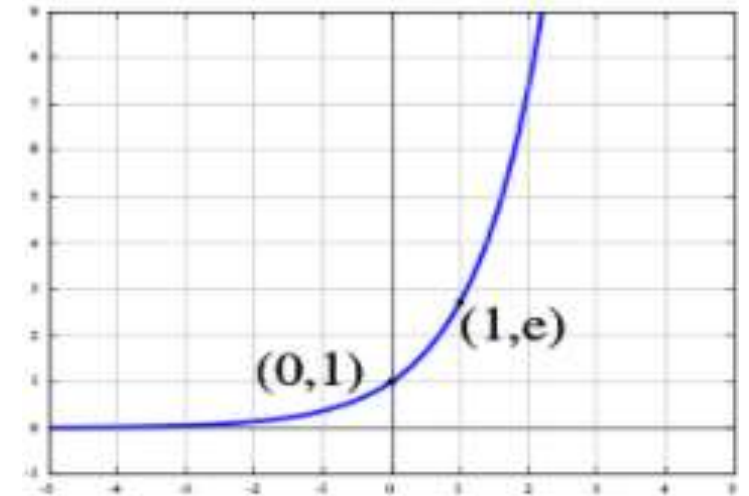


Multiclass Classification: Softmax

Numerical stability

$$p(y_i|\mathbf{x}_i; \Theta) = \frac{e^{s_{yi}}}{\sum_{k=1}^C e^{s_k}} = \frac{e^{s_{yi} - s_{\max}}}{\sum_{k=1}^C e^{s_k - s_{\max}}}$$

exponential function

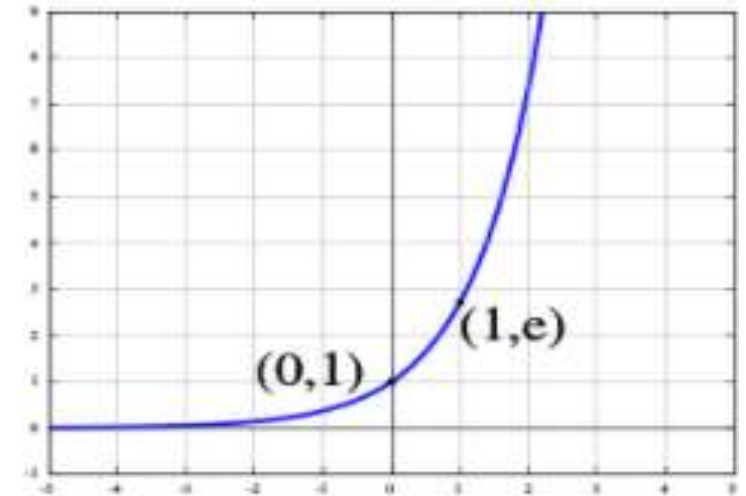


Multiclass Classification: Softmax

Numerical stability

$$p(y_i|\mathbf{x}_i; \Theta) = \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}} = \frac{e^{s_{y_i} - s_{\max}}}{\sum_{k=1}^C e^{s_k - s_{\max}}}$$

exponential function



Cross-Entropy Loss

$$L_i = -\log(p(y_i|\mathbf{x}_i; \Theta)) = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes



Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes



label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$$

Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes



label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$$

$$L_i = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$
 ↓
 scores

			
cat	3.2	1.3	2.2
chair	5.1	4.9	2.5
car	-1.7	2.0	-3.1

Loss

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

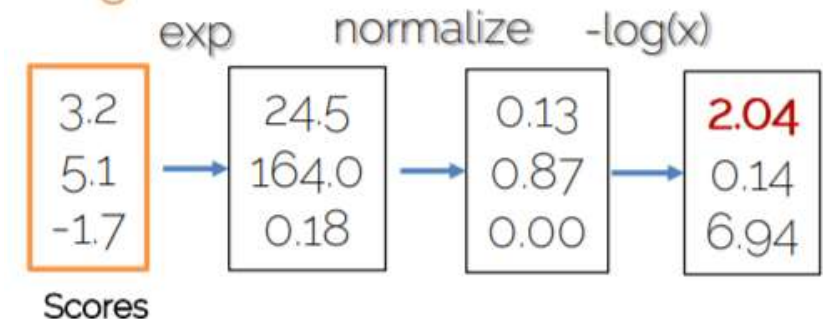
Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Image 1



Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.04		

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

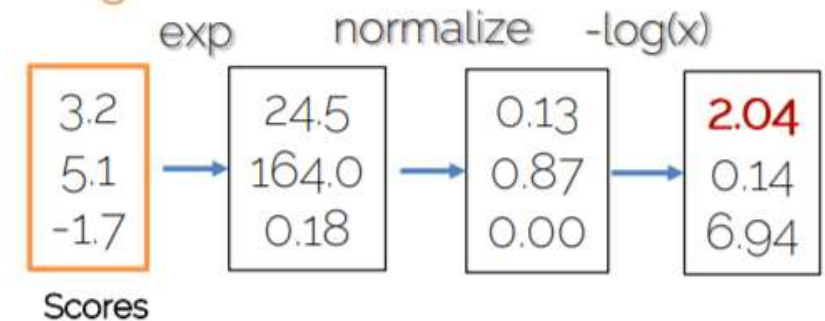
Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Image 1



Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.04	0.079	6.156

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.04	0.079	6.156

Cost function

$$L = \frac{1}{N} \sum_{i=1}^N L_i$$

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.04	0.079	6.156

Cost function

$$L = \frac{1}{N} \sum_{i=1}^N L_i = \frac{L_1 + L_2 + L_3}{3}$$

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Example: Cross-Entropy Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.04	0.079	6.156

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = -\log \frac{e^{s_{y_i}}}{\sum_{k=1}^C e^{s_k}}$$

Cost function

$$L = \frac{1}{N} \sum_{i=1}^N L_i = \frac{L_1 + L_2 + L_3}{3} = \frac{2.04 + 0.079 + 6.156}{3} = 2.76$$

Hinge loss

Score Function

$$\begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_C \end{bmatrix} = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix} \begin{bmatrix} x_{i0} \\ x_{i1} \\ \vdots \\ x_{id} \end{bmatrix}$$

Hinge loss

Score Function

$$\begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_C \end{bmatrix} = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix} \begin{bmatrix} x_{i0} \\ x_{i1} \\ \vdots \\ x_{id} \end{bmatrix}$$

Hinge Loss

$$L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

Example: Hinge Loss

Suppose: 3 training examples and 3 classes



Example: Hinge Loss

Suppose: 3 training examples and 3 classes



label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $\mathbf{x}_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\boldsymbol{\theta}_1^T & - \\ -\boldsymbol{\theta}_2^T & - \\ \vdots & \\ -\boldsymbol{\theta}_C^T & - \end{bmatrix}$$

Example: Hinge Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$

↓

scores

cat			
cat	3.2	1.3	2.2
chair	5.1	4.9	2.5
car	-1.7	2.0	-3.1

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

Example: Hinge Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

Example: Hinge Loss



Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

$$L_1 = \max(0, 5.1 - 3.2 + 1)$$

Example: Hinge Loss



Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

$$L_1 = \max(0, 5.1 - 3.2 + 1) + \max(0, -1.7 - 3.2 + 1)$$

Example: Hinge Loss



Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.9		

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

$$L_1 = \max(0, 5.1 - 3.2 + 1) + \max(0, -1.7 - 3.2 + 1) = 2.9 + 0 = 2.9$$

Example: Hinge Loss



Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.9	0	

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

$$L_2 = \max(0, 1.3 - 4.9 + 1) + \max(0, 2.0 - 4.9 + 1) = 0 + 0 = 0$$

Example: Hinge Loss



Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.9	0	12.9

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

$$L_3 = \max(0, 2.2 + 3.1 + 1) + \max(0, 2.5 + 3.1 + 1) = 6.3 + 6.6 = 12.9$$

Example: Hinge Loss

Suppose: 3 training examples and 3 classes

$s_k = \theta_k^T x_i$				
scores ↓	cat	3.2	1.3	2.2
	chair	5.1	4.9	2.5
	car	-1.7	2.0	-3.1
Loss		2.9	0	12.9

label (true class)
 $y_i \in \{1, 2, \dots, C\}$

Input vector
 $x_i \in \mathbb{R}^D$

Parameters

$$\Theta = \begin{bmatrix} -\theta_1^T & - \\ -\theta_2^T & - \\ \vdots & \\ -\theta_C^T & - \end{bmatrix}$$

$$L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

Cost function

$$L = \frac{1}{N} \sum_{i=1}^N L_i = \frac{L_1 + L_2 + L_3}{3} = \frac{2.9 + 0 + 12.9}{3} = 5.27$$

Example: Hinge vs Cross-Entropy

$$\text{Hinge Loss: } L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

$$\text{Cross Entropy: } L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_k e^{s_k}}\right)$$

For image $\mathbf{x}_i$ (assume $y_i = 0$):

	Scores	Hinge loss:	Cross Entropy loss:
Model 1	$s = [5, -3, 2]$		
Model 2	$s = [5, 10, 10]$		
Model 3	$s = [5, -20, -20]$		

Example: Hinge vs Cross-Entropy

$$\text{Hinge Loss: } L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

$$\text{Cross Entropy: } L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_k e^{s_k}}\right)$$

For image $\mathbf{x}_i$ (assume $y_i = 0$):

	Scores	Hinge loss:	Cross Entropy loss:
Model 1	$s = [5, -3, 2]$	$\max(0, -3 - 5 + 1) + \max(0, 2 - 5 + 1) = 0$	
Model 2	$s = [5, 10, 10]$	$\max(0, 10 - 5 + 1) + \max(0, 10 - 5 + 1) = 12$	
Model 3	$s = [5, -20, -20]$	$\max(0, -20 - 5 + 1) + \max(0, -20 - 5 + 1) = 0$	

Apparently Model 3 is better, but losses show no difference between Model 1&3 since they all have same loss=0.

Example: Hinge vs Cross-Entropy

$$\text{Hinge Loss: } L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$$

$$\text{Cross Entropy: } L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_k e^{s_k}}\right)$$

For image $\mathbf{x}_i$ (assume $y_i = 0$):

	Scores	Hinge loss:	Cross Entropy loss:
Model 1	$s = [5, -3, 2]$	$\max(0, -3 - 5 + 1) + \max(0, 2 - 5 + 1) = 0$	$-\ln\left(\frac{e^5}{e^5 + e^3 + e^2}\right) = 0.05$
Model 2	$s = [5, 10, 10]$	$\max(0, 10 - 5 + 1) + \max(0, 10 - 5 + 1) = 12$	$-\ln\left(\frac{e^5}{e^5 + e^{10} + e^{10}}\right) = 5.70$
Model 3	$s = [5, -20, -20]$	$\max(0, -20 - 5 + 1) + \max(0, -20 - 5 + 1) = 0$	$-\ln\left(\frac{e^5}{e^5 + e^{-20} + e^{-20}}\right) = 2 * 10^{-11}$

Model 3 has a clearly smaller loss

Example: Hinge vs Cross-Entropy

Hinge Loss: $L_i = \sum_{k \neq y_i} \max(0, s_k - s_{y_i} + 1)$

Cross Entropy : $L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_k e^{s_k}}\right)$

For image $\mathbf{x}_i$ (assume $y_i = 0$):

	Scores	Hinge loss:	Cross Entropy loss:
Model 1	$s = [5, -3, 2]$	$\max(0, -3 - 5 + 1) + \max(0, 2 - 5 + 1) = 0$	$-\ln\left(\frac{e^5}{e^5 + e^3 + e^2}\right) = 0.05$
Model 2	$s = [5, 10, 10]$	$\max(0, 10 - 5 + 1) + \max(0, 10 - 5 + 1) = 12$	$-\ln\left(\frac{e^5}{e^5 + e^{10} + e^{10}}\right) = 5.70$
Model 3	$s = [5, -20, -20]$	$\max(0, -20 - 5 + 1) + \max(0, -20 - 5 + 1) = 0$	$-\ln\left(\frac{e^5}{e^5 + e^{-20} + e^{-20}}\right) = 2 * 10^{-11}$

– Cross Entropy *always* wants to improve! (loss never 0)

– Hinge Loss saturates.